

Peter Fu

Mobile: +86 13735843901 | E-mail: fuchengdong1@outlook.com | Country: China

Education Background

The University of Sydney

Feb. 2022 – Feb. 2023

MSc. Data Science

Sydney, Australia

The University of Sydney

Mar. 2018 – Aug. 2021

BSc. Computer Science

Sydney, Australia

Summary

Hi! I'm Peter Fu, a Machine Learning Engineer with 3 years of industry experience in **Generative AI & Computer Vision**. My previous work focuses on **3D World Model**, **Video Generation**, and **Interactive Image Control**. I have **studied 100+ papers** from top conferences (CVPR, ICCV, NeurIPS, etc.), could quickly implementing solutions with rigorous **A/B testing** for performance validation. My approach emphasizes understanding models from first principles, which enables me to design custom modules tailored to specific business requirements and deliver practical solutions to real-world challenges.

Work Experience

ShuYueYun Tech. (Startup)

Aug. 2024 – Dec. 2025

Research Engineer

Hangzhou, China

- Implemented a **2B-Parameter World Model** to generate simulated videos of 3D indoor virtual scenes, enabling free-viewpoint rendering, prompt-guided scene control, and interactive navigation (forward/backward) for exploration and camera motion control.
- Built an interactive editing plates by **SAM 3D**, enabling users to segment and customize arbitrary indoor elements through intuitive click-and-edit interactions with **3D-Aware Segmentation**.
- Designed the physical relational loss with **Meta-Learning Optimization**, leveraging spatio-temporal alignment to learn effective physics commonsense across diverse scenarios, improving the Physical Commonsense (PC) score by 12.1% overall.
- Innovated a decoupled spatial attention module specifically tailored for Flow Matching Architectures (**Flux 1.**), incorporating spatial contrastive loss to enhance spatial coherence, resulting in 4.6% FID metric improvements on in-domain dataset.
- Optimized the **User-Customized Image & Video Generation** module by applying **Free Lunch methods** to **DiT** structures, increasing facial consistency precision by 5.7% on in-domain datasets.

MOGU

Aug. 2023 – July. 2025

Machine Learning Engineer

HangZhou, China

- Led end-to-end development of an **Interactive Image Generation Pipeline** for User-Customized services, including dataset creation, model construction, post-processing and evaluation. Deployed the model on WeShop-AI, enabling multi-modal editing through text prompts, reference images, masks, and bounding boxes for **Region-Specific Generation and Edition**.
- Performed **Large-Scale Distributed Pretraining** of **Stable Diffusion XL** model on 1M e-commerce product images using 8x H800 GPUs, achieving 21% improvement in FID score for domain-specific image generation.
- Optimized inference pipeline through **LLM-based Chain-of-Thought (CoT)** prompt engineering, enhancing semantic generation consistency with a 1.4% improvement in CLIP score.
- Collaborated with backend team on API service deployment, achieving **100K monthly API calls** while maintaining cost-efficient inference.

Skills

Computer Vision & Generative AI: image generation models (Flux, Stable Diffusion etc.), video generation models (Open-Sora, Stable Video Diffusion, Wan etc.), 3D Word models (LWM, HunyuanWorld etc.), Segmentation models (SAM, RMBG etc), Vision-Language models (CLIP, Grounding DINO etc.), LLM models (Llama, Mixtral etc.)

Machine Learning & Optimization: Transfer learning, Zero-shot learning, Meta learning, Mixed precision training

Languages: English (IELTS: 6.5), Mandarin (native)